Boundary conditions on non-planar boundaries

Louis Moresi [1]

[1] Australian National University

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A free surface moves under the traction it carries. That makes the wall-normal stress the quantity driving the model rather than something read off at the end, and it is the reason we went back to how the boundary condition underneath it is imposed.

The condition itself is ordinary. Free slip is no flow through the boundary and no tangential drag along it,

$$\boldsymbol{u} \cdot \hat{\boldsymbol{n}} = 0 \qquad \text{and} \qquad \hat{\boldsymbol{t}} \cdot \boldsymbol{\sigma} \hat{\boldsymbol{n}} = 0. \tag{1}$$

On a box the first of those is a single velocity component. You hold u_x on a vertical wall, the solver removes a row, and there is nothing further to discuss. On a sphere, an annulus, a mesh that has been moved, or a surface with topography, $\boldsymbol{u} \cdot \hat{\boldsymbol{n}}$ is not a component of anything. That is the whole difficulty, and everything below is a way around it.

The second condition is worth naming because it is the one people forget. Zero tangential traction is *natural*: it is what you get by leaving the boundary term out of the weak form. Nothing has to be done to impose it, and something has to be done to avoid imposing it accidentally.

1. Where the boundary term comes from

Every method below is a statement about one term. Multiplying the momentum balance by a test function $\boldsymbol{w}$ and integrating by parts gives

$$\int_\Omega \boldsymbol{\sigma} : \nabla \boldsymbol{w} \, \mathrm{d}V - \int_{\partial\Omega} (\boldsymbol{\sigma} \hat{\boldsymbol{n}}) \cdot \boldsymbol{w} \, \mathrm{d}S = \int_\Omega \boldsymbol{f} \cdot \boldsymbol{w} \, \mathrm{d}V. \tag{2}$$

Drop the surface integral and you have imposed zero traction in both directions — free *everything*, not free slip. Free slip keeps the tangential half of that and replaces the normal half with the constraint. How you do the replacing is the choice.

2. Three ways, in order

2.a. A direct penalty

Add a term that punishes any flow through the boundary:

$$\dots + \frac{\gamma}{h}\int_{\partial\Omega} (\boldsymbol{u}\cdot\hat{\boldsymbol{n}})(\boldsymbol{w}\cdot\hat{\boldsymbol{n}})\ \mathrm{d}S. \tag{3}$$

One line, no new machinery, and it works on any geometry. What you are solving is a perturbed problem, though, and it is perturbed by exactly the amount the constraint is violated: the discrete solution sits where the penalty term balances the traction it is fighting, which leaves $\boldsymbol{u}\cdot\hat{\boldsymbol{n}}$ small but not zero. Making it smaller means raising γ, and raising γ conditions the operator worse. The error is traded against the conditioning, and neither can be driven away.

2.b. Nitsche

The reason the penalty is only accurate in the limit is that it is not *consistent*: substituting the true solution does not leave the equation satisfied, because the true solution is subject to a boundary traction the penalty form ignores. Nitsche's method (Nitsche, 1971) restores consistency by carrying that traction explicitly:

$$\dots - \int_{\partial\Omega} (\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}(\boldsymbol{u})\hat{\boldsymbol{n}})(\boldsymbol{w}\cdot\hat{\boldsymbol{n}})\ \mathrm{d}S - \int_{\partial\Omega} (\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}(\boldsymbol{w})\hat{\boldsymbol{n}})(\boldsymbol{u}\cdot\hat{\boldsymbol{n}})\ \mathrm{d}S + \frac{\gamma}{h}\int_{\partial\Omega} (\boldsymbol{u}\cdot\hat{\boldsymbol{n}})(\boldsymbol{w}\cdot\hat{\boldsymbol{n}})\ \mathrm{d}S. \tag{4}$$

The first of the three is the consistency term: it is the boundary traction the integration by parts produced, and putting it back is what makes the true solution satisfy the discrete equations exactly. The second is its transpose, which keeps the form symmetric and buys optimal convergence in L^2. The third is the penalty again, and it is still needed — but now for *stability* rather than for accuracy, and γ has a threshold set by an inverse inequality rather than being a dial you turn up until the answer looks right.

This is a real improvement and it is still a weak imposition. The constraint holds to the accuracy of the discretisation, not to the accuracy of the arithmetic.

2.c. Rotating the degrees of freedom

Stop asking for the constraint and impose it. At each constrained node, change the basis in which the velocity unknowns are expressed, from the global Cartesian frame to the local $(\hat{\boldsymbol{n}}, \hat{\boldsymbol{t}})$ frame. In that basis "no flow through the boundary" is again a single component, and it is removed the same way it would be on a box.

Collect the per-node rotations into a block-diagonal Q, equal to the identity at every node that is not constrained. The rotated system is

$$\hat{A} = Q^T A Q, \qquad \hat{\boldsymbol{b}} = Q^T \boldsymbol{b}, \qquad \boldsymbol{u} = Q\hat{\boldsymbol{u}}, \tag{5}$$

and the wall-normal row of $\hat{A}$ is struck out. The constraint then holds to machine precision, because it is not being solved for at all.

This is the classical answer, not a new one. It is in the early finite-element literature, and Engelman et al. (1982) were already reviewing the alternatives and choosing between them on grounds of global mass conservation in 1982. What is worth explaining is not the idea but why, given that it is exact and the others are not, it is the least used of the three.

3. What it costs, and the cost is structural

Rotating the degrees of freedom leaves the discrete problem in a **mixed basis**. Interior nodes hold (u_x, u_y); constrained nodes hold (u_n, u_t). Nothing about that is difficult in itself, and everything downstream has to agree about which nodes are which.

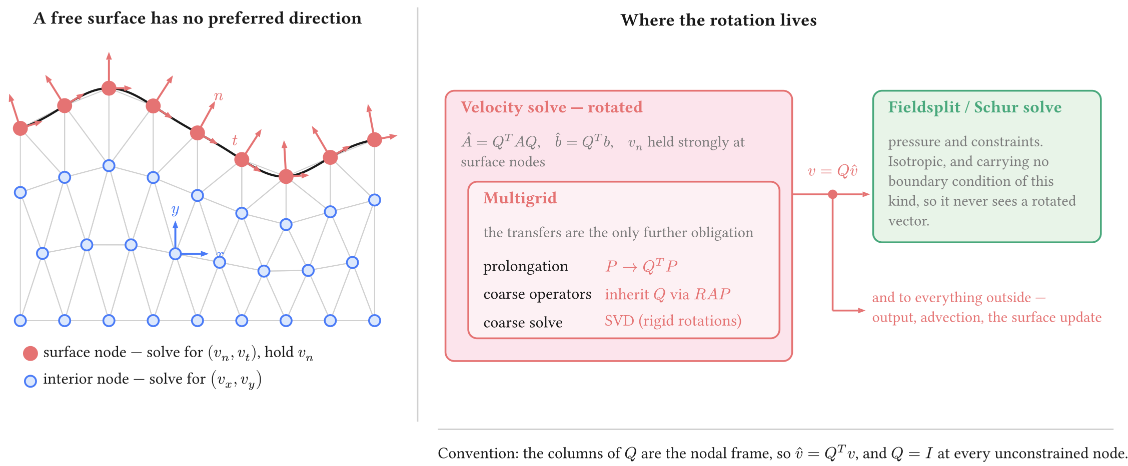

Figure 1: Where the rotation lives. The obligation is contained: the velocity solve is rotated and carries its multigrid with it, while the Schur complement and the pressure solve beside it never handle a rotated vector, because the pressure block carries no boundary condition of this kind. One un-rotation sits on the boundary between them and feeds both.

Four things carry Q: the operator, the right-hand side, the solution on the way out, and the multigrid prolongation. The coarse operators inherit it through the Galerkin triple product rather than being rotated separately, and the coarse solve then has to be an SVD, because a Galerkin-coarsened rotated operator inherits the rigid-rotation null space of the constrained problem and a redundant LU factorisation meets a zero pivot in it.

That last point is worth dwelling on, because it is the one that surprises. The constraint never has to be re-imposed on a coarse mesh — which is fortunate, since we install no discretisation there and could not impose it if we wanted to. It arrives anyway, algebraically, through $P^T \hat{A} P$ with a rotated P. The evidence that it really arrives is that the coarse operator inherits the null space, which is a property only the constrained problem has.

None of this is an argument against the method. It is an argument for knowing what is being taken on, and it is the honest reason a weakly imposed condition survives in codes that could do this instead.

4. Which normal

A question with a less obvious answer than it looks. The boundary of a discretised domain is a set of straight facets, and the assembled constraint is an integral over those facets. The node normal consistent with that integral is the average of the adjacent facet normals **weighted by facet measure** — not the normal of the smooth surface the mesh approximates.

Using the analytic normal is exact for the geometry and therefore inconsistent with the discretisation, which is the wrong way round: the solver is not solving on the sphere, it is solving on the polyhedron. In parallel the same argument has a sharper edge, because a normal accumulated rank-locally is wrong at a partition boundary, where a node's facets are split across ranks and no rank sees them all.

Worth checking against the literature

Behr (2004) addresses this question directly for slip conditions on curved boundaries. We have not read it against our own derivation. If it reaches the same measure-weighted normal, this section should say so and cite it as the source rather than presenting the result as ours.

5. The option this note leaves out

Solving in spherical or cylindrical components makes the wall-normal direction a coordinate direction again, and the constraint returns to being "hold one component". That is the same rotation as above, applied once for the whole domain instead of node by node, and applied that way it costs nothing structurally: there is no mixed basis, because every node is in the same basis.

It works exactly when the boundary lies along a coordinate surface. A sphere, an annulus, a cylinder. It does nothing for topography, for a mesh that has been deformed, or for a tilted internal surface, which is the general case this note is about. The per-node rotation is what remains once the geometry stops cooperating, and its structural price is what buys the generality.

6. When the choice matters

Here is the awkward part. Solve a convection model with any of the three and the velocity field is the same to plotting accuracy. A leak of order 10^{-3} in $\boldsymbol{u} \cdot \hat{\boldsymbol{n}}$ is invisible to anything that consumes the velocity, and consuming the velocity is most of what a model does. If that is your situation, use the simplest thing that works and stop reading here.

The difference appears when the wall-normal traction is the answer rather than a by-product: dynamic topography, a plate-boundary force balance, anything compared against a geoid or a gravity field. Then the three separate, and they separate for a structural reason rather than by a matter of degree.

Under a penalty or a Nitsche condition the constraint is approximate, so a traction recovered from the solution inherits the approximation — you are differentiating a field that was never made to satisfy the condition exactly. Under the rotated constraint the reaction *is* σ_{nn}: it is the multiplier the solve has already computed, and it comes out of `boundary_normal_traction` without a recovery step or an augmented-Lagrangian splitting.

This section is not yet measured

The argument above is structural and we believe it, which is not the same as having shown it. The note needs one experiment: the same model under all three boundary treatments, with the surface traction compared against a case whose answer is known. Until that table is here, this section is a claim.

7. Using it

```python
import underworld3 as uw

mesh = uw.meshing.Annulus(radiusInner=0.5, radiusOuter=1.0, cellSize=0.05)
stokes = uw.systems.Stokes(mesh)

# Value first: 0 is free slip. A non-zero scalar or expression prescribes the
# wall-normal datum u.n = u_n strongly instead.
stokes.add_rotated_freeslip_bc(0.0, "Upper")
stokes.add_rotated_freeslip_bc(0.0, "Lower")

stokes.solve()

# The constraint reaction, which is the boundary normal traction.
sigma_nn = stokes.boundary_normal_traction("Upper")
```

Leave the normal to Underworld unless the constraint has to follow the true surface rather than the mesh. Passing an analytic normal — `X / |X|` on a sphere — is exact for the geometry and keeps a consistency error against the faceted assembly, which is usually not what you want.

Reach for Nitsche when the boundary condition has to **change during the model**. A hard constraint cannot morph: a wall that begins as a prescribed velocity and relaxes to a prescribed traction is a Nitsche problem, because the rotated constraint is either imposed or it is not.

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